



King Fahd University of Petroleum & Minerals

College of Computing and Mathematics

Information and Computer Science Department

ICS 344: Information Security (242)

Phase 3:

Defensive Strategy – Fail2ban on Metasploitable3

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Objective

Protect the Metasploitable3 SSH service from brute-force attacks using Fail2ban

Setup

- Installed and configured Fail2ban.
- Edited jail.local to monitor /var/log/auth.log.
- Jail [sshd] activated with:
maxretry = 2
bantime = 600s
- Restarted the service after configuration.

Testing

- Repeated failed logins from another machine (Mac terminal).
- Confirmed defense using:
fail2ban-client status sshd
Log files (/var/log/fail2ban.log and /var/log/auth.log)
- Verified automatic IP banning after exceeding threshold which is 3 attempts.

Problems Faced

- VM instability due to disk space and configuration.
- Mistakes in jail.local syntax.
- Initial failure to detect bans caused by message compression in logs.
- Manually tested and fixed by forcing bans and disabling message reduction.

Outcome

- SSH brute-force detection confirmed.
- IP banning observed and logged.
- Fail2ban working as expected.